

Yashraj Kastode

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Education

Integrated M.Tech – Artificial Intelligence

VIT Bhopal University, Bhopal

2022 – 2027

CGPA: 8.53

Technical Skills

Programming: C++, Python, JavaScript

Core CS: Data Structures & Algorithms, OOPS, DBMS

Backend / Web: Django, Django REST Framework

Tools / Platforms: Git, GitHub, Redis

ML / Libraries: NumPy, Pandas, TensorFlow, PyTorch, Hugging Face Transformers

Projects

Hindi AI Text Summarizer | Abstractive NLP System

[Link](#)

(*mT5, Transformers, Django REST, Celery, Redis*)

- Developed abstractive Hindi summarization system using **google/mt5-base Transformer**, processing 500+ samples.
- Implemented scalable inference pipeline with **Django REST Framework**, enabling real-time NLP predictions.
- Integrated **Celery + Redis** for asynchronous task execution, reducing latency by 40%.
- Improved output coherence using **beam search decoding**, reducing repetition errors by 30%.

Med-Matrix | AI-powered Drug Interaction Analysis System

[Link](#)

(*NLP Concepts, Knowledge Representation, Django REST, Redis*)

- Designed an **AI-assisted drug interaction system** using structured pharmaceutical data and rule-based inference.
- Modeled drug compositions via **salt-level semantic mapping**, enabling interaction detection across 100+ medicines and 300+ salts.
- Built an **interaction reasoning engine** to identify conflicting drug combinations using domain-specific logic.
- Developed scalable APIs with **Django REST Framework** for real-time query resolution.
- Optimized performance using **Redis caching / async processing**, reducing lookup latency by 60%.

Hasthakshar | Indian Sign Language Recognition

[Link](#)

(*Computer Vision, MediaPipe, OpenCV, TensorFlow, PyTorch*)

- Built dataset of 1000+ labeled samples for Indian Sign Language gesture recognition.
- Extracted skeletal keypoints using **MediaPipe** and optimized preprocessing pipeline with OpenCV.
- Trained deep learning classification model achieving **90% accuracy**.
- Optimized inference pipeline to achieve **15+ FPS** for near real-time gesture recognition.

Achievements

- **Microsoft Office Specialist: PowerPoint 2013 (Microsoft Certified)** – May 2019 [Badge Link]

Leadership & Extracurriculars

- **President & Founder**, Eureka Club – VIT Bhopal
- **Volunteer**, AI Conclave 2.0 – AI Club, VIT Bhopal